

Leaky Images: Targeted Privacy Attacks in the Web

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Leaky Images: Targeted Privacy Attacks in the Web

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Abstract:

Sharing files with specific users is a popular service provided by various widely used websites, e.g., Facebook, Twitter, Google, and Dropbox. A common way to ensure that a shared file can only be accessed by a specific user is to authenticate the user upon a request for the file. This paper shows a novel way of abusing shared image files for targeted privacy attacks. In our attack, called leaky images, an image shared with a particular user reveals whether the user is visiting a specific website. The basic idea is simple yet effective: an attacker-controlled website requests a privately shared image, which will succeed only for the targeted user whose browser is logged into the website through which the image was shared. In addition to targeted privacy attacks aimed at single users, we discuss variants of the attack that allow an attacker to track a group of users and to link user identities across different sites. Leaky images require neither JavaScript nor CSS, exposing even privacy-aware users, who disable scripts in their browser, to the leak. Studying the most popular websites shows that the privacy leak affects at least eight of the 30 most popular websites that allow sharing of images between users, including the three most popular of all sites. We disclosed the problem to the affected sites, and most of them have been fixing the privacy leak

in reaction to our reports. In particular, the two most popular affected sites, Facebook and Twitter, have already fixed the leaky images problem. To avoid leaky images, we discuss potential mitigation techniques that address the problem at the level of the browser and of the image sharing website.

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